diverted to the outlet, which opens after the high-pressure pulse passes. At high flow rates, the time of transition between the two states (which is a measure of the storage device time delay) is around 100 ms. The stability of the narrow stream from inlet 2 is because of the large extensional stresses, resembling the phenomenon of an "open siphon" (*6*, *14*), well known in polymer solutions.

The difference in fluxes between the two states can be converted to a difference in pressures by putting fluidic resistors (long, narrow channels) between the active element exits and the outlets, Fig. 3B. The pressure modulation between the two states is shown in Fig. 4B as a function of the total maximal pressure difference. The pressure modulation is more than 50% at highest ΔP, reaching 1800 Pa.

The transition to the functional nonlinear regime in these devices is determined by $Wi = \lambda \nabla V$, whereas the driving pressure and pressure modulation are proportional to η/λ. Therefore, the characteristic pressures should remain constant when the devices are scaled down to dimensions of a few tens of micrometers. We verified that by building functional flip-flops with four-times-smaller channel depths and active element areas. Further, by using a lower molecular weight polymer sample with smaller λ, we can increase the characteristic fluxes and pressure modulations while the response time is reduced. The use of smaller polymer molecules should also allow further miniaturization of the fluidic devices. So, we believe that a reduction in size of the devices described in this report by a factor of at least 10 is quite feasible.

We have observed purely elastic flow transitions and elasticity-generated flow patterns in dilute aqueous polymer solutions in microchannels of sub-100-μm width and demonstrated that those nonlinear flow phenomena can be used to construct functional microfluidic memory and control devices, which can work at arbitrarily low *Re* and are insensitive to electromagnetic noise. The pressure modulation achieved in the flip-flop is nearly sufficient to actuate other functional elements such as membrane valves (*17*), thus opening the possibility for fabrication of cascaded devices and multielement control circuits. If a biocompatible polymer solution is used, the flux stabilizer could find medical applications for implanted drug-delivery devices, for which a constant flux of the pharmaceutical into the patient is required (*18*).

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9. Microchannels were fabricated by molding an optically transparent silicon elastomer, polydimethylsiloxane (PDMS). Molds were prepared by spin coating a silicon wafer with a layer about 100 μm thick of an ultraviolet (UV)-curable epoxy (MicroChem SU8-50) and exposing it to UV light through a photomask, which had a resolution of 7.1 μm. The mold was used to make a 4-mm-thick cast of PDMS (RTV 615 by General Electric) with 100-μm-deep rectangular grooves on its surface. Finally, holes were punched through the elastomer for feeding the liquids, and the side with the grooves was bonded to a microscope coverslip by overnight baking in an 80°C oven. To fabricate the pressure sensors for the device in Fig. 3B, we performed two consecutive photolithography steps in mold fabrication; the first used a 5-μm-thick coating of SU8-2005, whereas the second was as described above.
10. A 250 parts per million by weight solution of a high molecular weight (M_w) polyacrylamide, $M_w = 1.8 \times 10^7$, from Polysciences (Warrington, PA) was used. The Newtonian solvent was an aqueous solution of 13% sucrose, 1% NaCl, and 0.1% Tween (Sigma) surfactant. The sucrose was added to bring the solution density to 1.055 g cm^{-3} and to ensure neutral buoyancy of suspended polystyrene beads, and salt was added to fix the ionic contents. The solvent viscosity, η_s, was 1.37×10^{-3} Pa · s at the room temperature of 22°C. The solution viscosity was 2.78×10^{-3}Pa · s at a shear rate of 150 s^{-1}, suggesting a dilute-to-semidilute polymer solution. The relaxation time, λ, was estimated as 13 ms from the relaxation-time measurements of a solution of the same polymer in a more viscous sugar syrup (*8*), with the assumption that λ scales linearly with η_s.
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13. The pressures were generated hydrostatically with the use of two long vertical rails with precise rulers and sliding stages. Working liquids were kept in three 30-ml plastic syringes to minimize effects of surface tension and level variation during the experimental runs. The syringes were held upright, open to the atmosphere, and connected to the inlets and the outlet by plastic tubing with an internal diameter of 0.76 mm. The pressure drop in the tubing was estimated to be below 1% of the total. The two syringes feeding the inlets were attached to the sliding stages. The difference in liquid elevation between these two syringes and the outlet syringe was measured with a precision of about 0.1 mm, corresponding to 1 Pa in pressure. During the experiment, the polymer solution was driven from inlet 1 (Fig. 1A) through the nonlinear channel a at varying pressure P_p. Pure solvent seeded with 1-μm beads at a high concentration was pumped through channel b from inlet 2 at pressure P_s. The two streams merged in region c with a sharp and laminar separation line (Fig. 1D). P_s was always tuned to bring the separation line to the same position at the center of channel c. After the dependence of P_s on P_p was found, polymer solution was pumped again through channel a at varying *Q* with the use of a syringe pump. Again, P_s was tuned to put the separation line at the center, and a smooth curve connecting *Q* and P_s was obtained. Thus, we could calculate the dependence of *Q* on P_p with an estimated relative error of about 0.5% without any direct measurements of the flow velocity. The error is mostly due to uncertainty in the tuning of P_s.
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18. This work was supported in part by the Defense Advanced Research Projects Agency, by a North Atlantic Treaty Organization postdoctoral fellowship to M.E., and by a Rothschild fellowship to A.G.

20 February 2003; accepted 4 April 2003

A Mesoscale Iron Enrichment in the Western Subarctic Pacific Induces a Large Centric Diatom Bloom

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We have performed an in situ test of the iron limitation hypothesis in the subarctic North Pacific Ocean. A single enrichment of dissolved iron caused a large increase in phytoplankton standing stock and decreases in macronutrients and dissolved carbon dioxide. The dominant phytoplankton species shifted after the iron addition from pennate diatoms to a centric diatom, *Chaetoceros debilis*, that showed a very high growth rate, 2.6 doublings per day. We conclude that the bioavailability of iron regulates the magnitude of the phytoplankton biomass and the key phytoplankton species that determine the biogeochemical sensitivity to iron supply of high-nitrate, low-chlorophyll waters.

Low iron bioavailability is thought to be one of the causes of high-nitrate and low-chlorophyll (HNLC) ocean waters (*1*, *2*) and may play a major role in linking global climate change and ocean productivity (*3*, *4*). Recent mesoscale iron enrichment experiments conducted in the equa-

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torial Pacific Ocean (IronEx I and II) (*2*, *5*) and the Southern Ocean (SOIREE and SOFeX) (*6*, *7*) confirmed that this iron limitation hypothesis is correct in these regions. The subarctic Pacific, with different biology (*8*) and strong longitudinal gradients in atmospheric iron deposition (*9*), has been the only major HNLC region where an iron enrichment experiment has not been performed.

We conducted a mesoscale, in situ, iron enrichment experiment, the Subarctic Pacific Iron Experiment for Ecosystem Dynamics Study (SEEDS), in the western subarctic gyre of the North Pacific (48.5°N, 165°E). Our experiment consisted of a single addition of 350 kg of iron as $FeSO_4$ with 0.48 M the inert tracer gas sulfur hexafluoride (SF_6), over an 8-km by 10-km patch with a mixed-layer depth of ~10 m, on 18 July 2001 (*10*). During the first survey on day 1 (19 July 2001), we observed a substantial increase in the dissolved iron concentration (2.9 nM, mean value) (Figs. 1B and 2B) in the patch in the surface mixed layer. The iron patch moved westward at ~6 km/day and was successfully traced for 2 weeks by tracking SF_6 (Figs. 1A and 2A).

A detectable increase in the photochemical quantum efficiencies of algal photosystem II [(F_v/F_m, where F_v is the difference between the maximum (F_m) and minimum (F_0) chlorophyl fluorescence yield] measurements by a fast repetition rate fluorometer (*11*) occurred after day 3 (Fig. 2F). The phytoplankton chlorophyll a in the patch showed a rapid and exponential increase from day 6, approaching 20 mg m^{-3} on day 10 (Figs. 1E and 2E). The phytoplankton biomass was relatively constant from day 9 to the end of the observation (day 13). The microsized (>10 μm in diameter) chlorophyll a comprised 36% of the total chlorophyll a before the iron enrichment and increased to 95% of the total chlorophyll a on day 13 (Fig. 3). Concentrations of nanosized (2 to 10 μm) and picosized (<2 μm) chlorophyll a also increased, but only by a factor of 2 to 5.

The phytoplankton that dominated the biomass in the iron-enriched patch changed from the pennate diatom *Pseudonitzschia turgidula* to chain-forming centric diatoms with one dominant species, *Chaetoceros debilis* (10 to 30 μm wide), in contrast to experiments in the equatorial Pacific (*2*, *5*) and the Southern Ocean (*6*), where iron stimulated the growth of pennate diatoms. *C. debilis* was a negligible component before the iron enrichment (240 cells l^{-1}), but became the most dominant species from day 6 on and finally accounted for 90% of microsized (>20 μm) cell density (Fig. 3). The net growth rate of *C. debilis* was as high as 2.6 doublings per day during days 4 to 7, which was much higher than the maximum growth rate expected for diatoms at an in situ temperature of 9.5°C [~1.5 doublings per day (*12*)].

The large increase in phytoplankton standing stock was accompanied by large decreases in the concentrations of macronutrients (nitrate, phosphate, and silicate), CO_2 fugacity (f_{CO_2}), and dissolved inorganic carbon (DIC) (Figs. 1, C and D; 2, C and D; and table S1), to a greater extent than previous iron-enrichment experiments (*2*, *5*, *6*). The maximum differences between f_{CO_2} and nitrate concentration in the surface water in and out of the patch were observed on day 12 (94 μatm and 9.0 μM, respectively).

F_v/F_m peaked on day 9 and then decreased (Fig. 2F), which suggests that the phytoplankton were physiologically stressed after day 9. Macronutrients decreased but were not depleted at the end of observation. Meanwhile, the dissolved iron concentration was ~0.15 nM, 25% of the half-saturation constant of the iron-growth relationship for the ambient phytoplankton community (*13*). These results indicate that phytoplankton in the patch became iron-stressed.

A relatively high export flux of carbon in the patch was observed between day 10 and 12; however, the increase in the patch was not statistically significant (table S1). The export flux between day 4 and day 13 was 11% of the integrated primary production in the patch (table S1). Moreover, the increase in particulate organic carbon in the surface mixed layer was 78% of the sum of the decrease in DIC and the influx of CO_2 by air-sea exchange (fig. S1). These results show that the major part of the fixed carbon stayed in the surface mixed layer as particulate matter at the end of our observation. However, the decrease in F_v/F_m suggests that phytoplankton were approaching the end of the bloom. The dominant species *C. debilis* produces resting spores that have a high silicate content (*14*). Massive sinking of centric diatom cells or resting spores at the end of a spring bloom is a well-known phenomenon in the coastal subarctic Pacific (*15*). Similarities in the magnitude of phytoplankton cell numbers and species composition between an iron-induced open-ocean bloom and a spring bloom in coastal waters

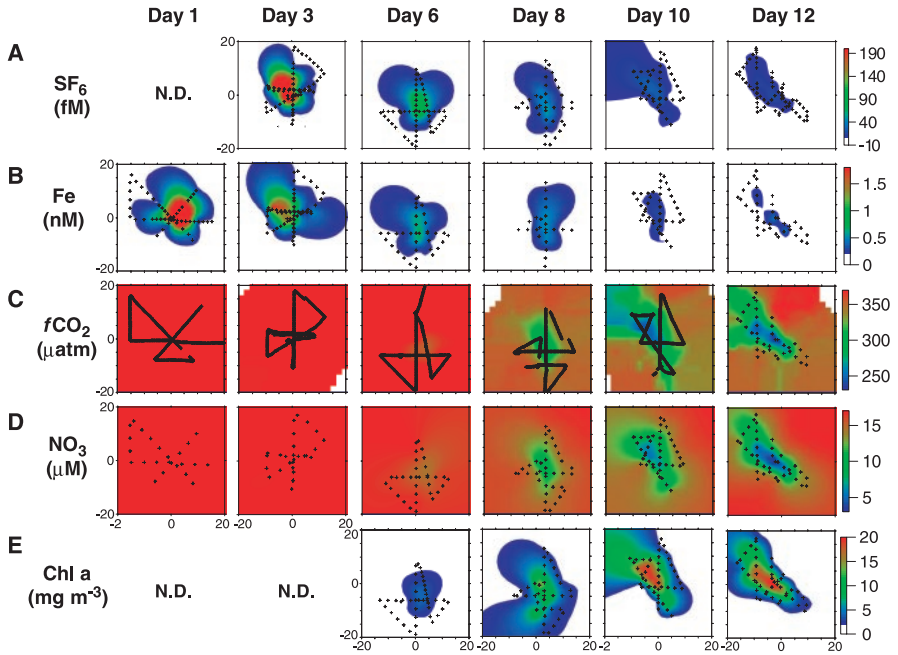

Fig. 1. Underway (at 6-m depth) maps of (**A**) SF_6 concentrations, (**B**) dissolved iron concentrations, (**C**) CO_2 fugacity (f_{CO_2}), (**D**) nitrate concentration, and (**E**) chlorophyll a (Chl a) concentrations during the daily mapping of the iron-enriched patch on day 1 (19 July), day 3 (21 July), day 6 (24 July), day 8 (26 July), day 10 (28 July), and day 12 (30 July). The plots denote the vessel's track relative to the center buoy during each survey. The *x* and *y* axes represent the distance (in kilometers) from the center buoy, and the concentration is indicated by the color scale on the right. N.D., no data.

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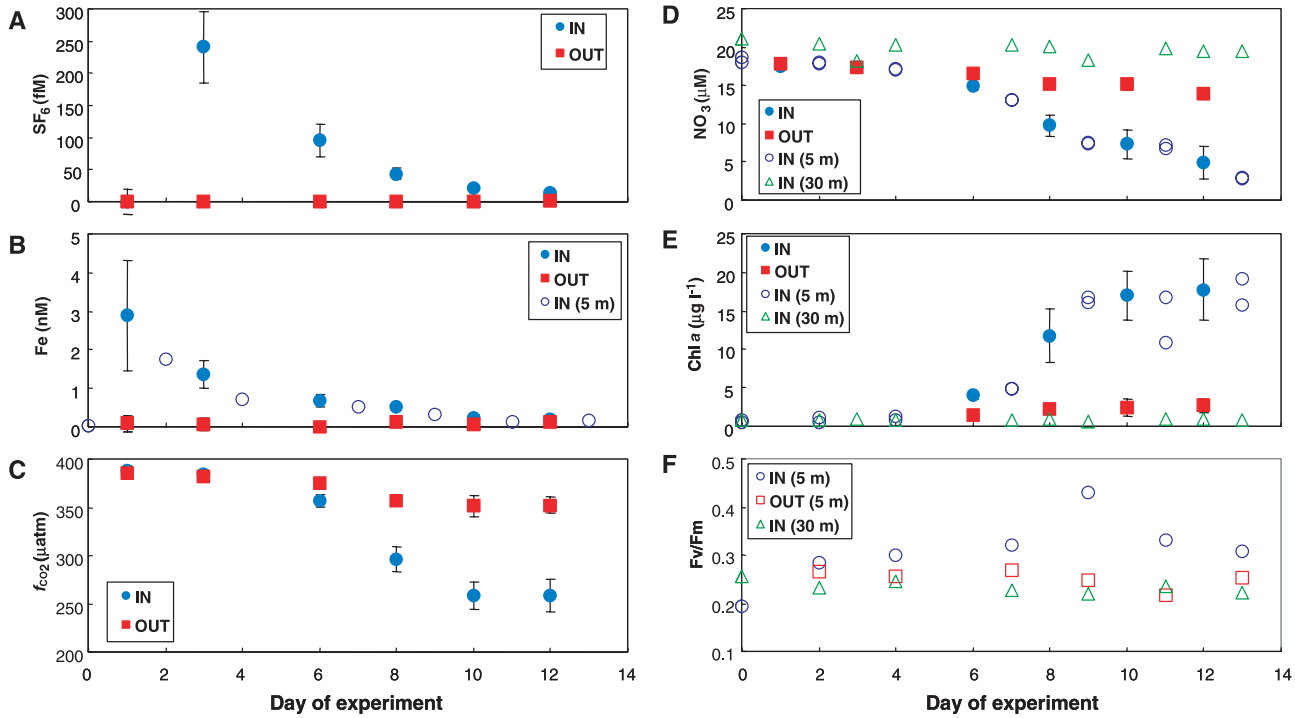

Fig. 2. Time series of six measured variables over the course of SEEDS. Solid symbols are underway data expressed as the daily mean value for inside (defined as >50% of the peak SF_6 levels on that day) and outside (defined as $SF_6 < 3$ fM) the patch. Error estimates are expressed as 1 standard deviation of the mean for underway data. Open symbols show discrete measurements made inside (at 5- and 30-m depth) and outside (at 5-m depth) the iron-enriched patch. Range of sample sizes: (**A**) SF_6 concentrations, $n = 9$ to 21 (in), 18 to 32 (out). (**B**) Dissolved iron concentrations, $n = 4$ to 13 (in), 6 to 23 (out). (**C**) CO_2 fugacity (f_{CO_2}), $n = 44$ to 118 (in), 117 to 296 (out). (**D**) Nitrate concentrations, $n = 4$ to 13 (in), 10 to 17 (out). (**E**) Chlorophyll a concentrations, $n = 5$ to 11 (in), 12 to 17 (out). (**F**) Community photosynthetic competency (F_v/F_m), $n = 3$.

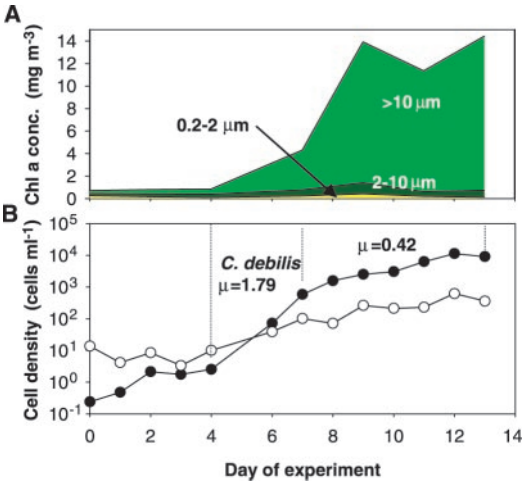

Fig. 3. Temporal changes of (**A**) size-fractionated chlorophyll a concentration and (**B**) cell densities of dominant phytoplankton species at 5-m depth in the iron-enriched patch. Day 0 indicates the day before fertilization. Solid circles, *C. debilis*; open circles, *P. turgidula*. Specific growth rates per day (µ) of *C. debilis* between day 4 and 7 and between 8 and 13 are given.

indicate that a substantial diatom sinking event might have occurred after our observation period.

The findings of SEEDS clearly demonstrate that iron availability controls phytoplankton abundance, thereby regulating biogeochemical processes in high-nutrient areas of the western subarctic Pacific that are similar to other HNLC waters. However, it also shows that all HNLC waters do not respond the same way to iron enrichment. The rapid and extensive response of phytoplankton observed in SEEDS is due to the presence of the fast-growing centric diatom *C. debilis*; the net growth rate of *C. debilis* was 1.8 times higher than that of pennate diatoms observed in IronEx II (*16*) and net algal growth rate in SOIREE (*6*). This indicates that the phytoplankton species composition of the pelagic ecosystem strongly influences the ecosystem's response to an increase in iron supply.

Martin's hypothesis (*17*) postulated that the increased iron supply during the Last Glacial Maximum relieved iron limitation of phytoplankton production in the Southern Ocean and that the corresponding increase in carbon export could explain the low atmospheric CO_2 levels during that period. Although the responses of the phytoplankton communities seen in iron-enrichment experiments have been transient, observations of SEEDS and the other experiments (*2*, *5*, *6*) support the former part of Martin's idea. The more valuable suggestion of our results, however, is the importance of phytoplankton species composition in determining the growth rate of diatoms, aggregation processes (*18*), and export flux during glacial periods.

Considering that the annual atmospheric iron fluxes in the SEEDS site are 10 to 100 times higher than those in the other HNLC areas (*9*), as well as the relatively high iron concentration in the mesopelagic layer in the western subarctic Pacific (*19*, *20*), a seed population of the fast-growing diatom *C. debilis* may be sustained by a sporadic supply of iron from dust (*21*) and from below the surface mixed layer. Such high aeolian iron depositions also occurred in the Southern Ocean during glacial periods (*22*, *23*). Therefore, we would expect a dominance of fast-growing centric diatoms, rather than moderate-growing pennate diatoms, in the glacial Southern Ocean. A high community-growth rate and the resultant accumulation of phytoplankton cells in the surface mixed layer promote the algal aggregation that is required for a massive and rapid sedimentation of organic particles (*18*). Thus, the possible floristic shift to fast-growing centric diatoms would accelerate a positive feedback effect on dust-iron-climate system.

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24. We thank P. J. Harrison, Y. W. Watanabe, H. J. W. de Baar, Y. Yokouchi, Y. Oozeki, C. S. Law, A. J. Watson, N. Handa, and the captain and crew of the Fisheries Research Vessel Kaiyo-Maru. Supported by the Global Environmental Research Fund from the Ministry of Environment, the Fisheries Agency, and Central Research Institute of Electric Power Industry research funding.

Supporting Online Material
www.sciencemag.org/cgi/content/full/300/5621/958/DC1
Materials and Methods
Fig. S1
Table S1

31 December 2002; accepted 4 April 2003

Sediment-Dispersed Extraterrestrial Chromite Traces a Major Asteroid Disruption Event

Birger Schmitz,[1,2]* Therese Häggström,[2] Mario Tassinari[2]

Abundant extraterrestrial chromite grains from decomposed meteorites occur in middle Ordovician (480 million years ago) marine limestone over an area of ~250,000 square kilometers in southern Sweden. The chromite anomaly gives support for an increase of two orders of magnitude in the influx of meteorites to Earth during the mid-Ordovician, as previously indicated by fossil meteorites. Extraterrestrial chromite grains in mid-Ordovician limestone can be used to constrain in detail the temporal variations in flux of extraterrestrial matter after one of the largest asteroid disruption events in the asteroid belt in late solar-system history.

More than 40 fossil meteorites (1 to 20 cm in diameter) were recovered in mid-Ordovician marine limestone in the Thorsberg quarry at Kinnekulle, southern Sweden (*1*). The high abundance of meteorites deposited in ancient sea-floor sediments indicates that the meteorite paleoflux was enhanced by up to two orders of magnitude as compared to the recent flux (*1*, *2*). Argon-isotope gas-retention ages of recent meteorites show that a major asteroid disruption event affecting the L chondrite parent body occurred in the asteroid belt at about 500 million years ago (Ma) (*3*, *4*). Relict chromite grains with an L chondritic composition in most of the mid-Ordovician fossil meteorites indicate that the meteorites represent the flux of objects to Earth shortly after the break up of the L chondrite body (*1*). We searched for dispersed extraterrestrial chromite grains from fossil meteorites in Ordovician limestones from other sites in Sweden to determine the extent and paleoflux of meteorites and to infer the size of the asteroid breakup event.

Recent meteorites of the ordinary chondritic group contain typically 0.05 to 0.5% of chromite ($FeCr_2O_4$) (*5*, *6*). This chromite has a characteristic composition, with higher Ti and lower Al and Mg contents than that of terrestrial chromite (*1*, *7*, *8*). Equilibrated meteorites of the three groups of ordinary chondrites show small differences in chromite compositions, with Fe and Ti contents increasing from H to L to LL chondrites (*1*, *7*). Chromite is a very resistant mineral that survives most kinds of acid treatment, and it is also one of the few relict minerals that occurs in mid-Ordovician fossil meteorites (*1*, *8*).

Large samples, 7 to 45 kg, of lower to middle Ordovician Orthoceratite Limestone (*9*) were searched for sand-sized chromite grains. The samples come from five sites in a 500-km transect in southern Sweden: Rävanäs, Hällekis, Thorsberg quarry, Gullhögen, and Degerhamn (Fig. 1). The Orthoceratite Limestone formed at average sedimentation rates of one to a few mm per thousand years in a large epicontinental sea covering large parts of Baltoscandia (*9*). The limestone is characterized by abundant hardgrounds, each representing a period of some 100 to 1000 years when weak currents in the water column prevented fine-grained detritus from settling on the sea floor (*10*).

We have focused the chromite search on the stratigraphic interval from which fossil meteorites have been recovered in the Thorsberg quarry (Fig. 2). The quarried 3.2-m thick interval is divided by the quarry workers into 12 beds with traditional names (*1*). The quarried interval represents circa (ca.) 1 to 2 million years (My) of sedimentation. The Arkeologen bed is the most meteorite-rich bed, with 26 meteorites found over a sea-floor area of 2700 m^2. The overlying Botten and Sextummen beds have yielded four and six meteorites, respectively, over 6000 m^2 of sea floor. The total recovered fossil meteoritic mass per unit sea-floor area in the three beds is 0.81, 0.69, and 0.17 kg 1000 m^{-2}, respectively. The Thorsberg quarry spans over three conodont zones: the lowermost *Lenodus antivariabilis* Zone, the middle *L. variabilis* Zone, and the upper *Yangtzeplacognathus crassus* Zone (Fig. 2) (*11*). The beds just below the Arkeologen bed belong to the *L. antivariabilis* Zone, and the beds just above the quarried interval belong to the *Microzarkodina hagetiana* Subzone.

From the Thorsberg quarry, we searched for chromite in limestone samples from the Arkeologen, Golvsten, Botten, and Sextummen beds (Fig. 2). The samples were collected at spots in the quarry where no fossil

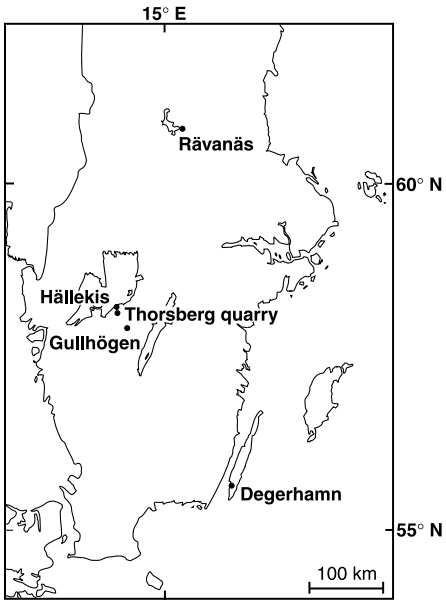

Fig. 1. Map of southern Sweden with sites studied.

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Science **300** (5621), . DOI: 10.1126/science.1082000

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